

CS401: Distributed Systems - Unit 2 Lecture Notes

Section 1: Raft Consensus Protocol and State Machine Replication

In distributed systems, consensus is the process of agreeing on a shared state among N nodes.

Raft accomplishes consensus through an elected leader that manages log replication.

The leader accepts log entries from clients, replicates them to follower nodes, and commits them.

To prevent split-brain anomalies during network partitions, a quorum of $N/2 + 1$ is required.

Section 2: Leader Election and Randomized Election Timeouts

When followers do not receive heartbeats within an election timeout, they transition to Candidate. The candidate increments its current term, votes for itself, and broadcasts RequestVote RPCs. Randomized election timeouts between 150ms and 300ms prevent split vote deadlocks. Once a majority quorum grants votes, the candidate becomes Leader and broadcasts heartbeats.